Appendix A: Tables of results

Table 1: Host star identifiers and coordinates.

Host star and KIC aperture	R.A.	Declination	Alias
Gaia DR3 2100980735718445312 (KIC 4459924)	290.23681871	39.56477993	2MASS J19205684+3933534
Gaia DR3 2100325529865507712 (KIC 4543171)	285.24349033	39.66402000	2MASS J19005844+3939502
Gaia DR3 2078315162505745920 (KIC 8175131)	295.00242100	44.09056099	2MASS 19400058+4405261
Gaia DR3 2126910621514995968 (KIC 8293539)	289.59770917	44.27846129	2MASS J19182345+4416426
Gaia DR3 2079369937757906688 (KIC 9040849)	297.60395913	45.36098508	2MASS J19502495+4521394
Gaia DR3 2085335681690420608 (KIC 9674230)	300.31550299	46.31674844	2MASS J20011572+4619005
Gaia DR3 2128959007378441472 (KIC 11033410)	293.63348302	48.55475499	2MASS J19343202+4833173

Table 2: Planetary parameters derived from PyTransit.

Host star and KIC aperture		b	k	u	v	a	i	$t14$	$t23$
Gaia DR3 2100980735718445312	Mean	0.6611	0.0752	0.3565	0.4636	6.0611	1.4614	0.0450	0.0342
(KIC 4459924)	Std. dev.	0.0389	0.0015	0.2030	0.3089	0.2157	0.0103	0.0003	0.0011
Gaia DR3 2100325529865507712	Mean	0.8087	0.0887	0.3695	0.3282	5.3011	1.4176	0.0444	0.0255
(KIC 4543171)	Std. dev.	0.0107	0.0015	0.2714	0.3346	0.0880	0.0044	0.0003	0.0010
Gaia DR3 2078315162505745920	Mean	0.6123	0.1349	0.5339	0.2482	9.4639	1.5060	0.0323	0.0203
(KIC 8175131)	Std. dev.	0.0825	0.0043	0.2392	0.3163	0.5444	0.0122	0.0005	0.0019
Gaia DR3 2126910621514995968	Mean	0.6490	0.1373	0.2842	0.5617	7.7727	1.4872	0.0385	0.0233
(KIC 8293539)	Std. dev.	0.0245	0.0018	0.1727	0.2565	0.1490	0.0046	0.0003	0.0009
Gaia DR3 2079369937757906688	Mean	0.6594	0.0772	0.4806	0.3318	6.8480	1.4744	0.0400	0.0300
(KIC 9040849)	Std. dev.	0.0978	0.0025	0.2515	0.3355	0.5795	0.0207	0.0008	0.0015
Gaia DR3 2085335681690420608	Mean	0.8022	0.1037	0.6094	0.0893	7.0327	1.4565	0.0349	0.0182
(KIC 9674230)	Std. dev.	0.0674	0.0038	0.3746	0.3823	0.5827	0.0162	0.0010	0.0031
Gaia DR3 2128959007378441472	Mean	0.7763	0.1146	0.3738	0.3129	5.6436	1.4328	0.0457	0.0242
(KIC 11033410)	Std. dev.	0.0179	0.0019	0.2655	0.3518	0.1226	0.0060	0.0005	0.0016

Notes. For each exoplanet candidate the mean sample value for each parameter is shown along with the standard deviation of the parameter value. The column headings have the following meanings:
b: impact parameter.
k: ratio of planet radius to stellar radius.
u and v: quadratic-model limb darkening coefficients.
a: ratio of orbital radius to stellar radius.
i: orbital inclination in radians.
$t14$: total duration of the transit in fractions of the orbital period.
$t23$: duration of the full transit in fractions of the orbital period.
NOTE: In all cases the orbital eccentricity was found to be negligibly small (of the order of 10^{-10}) and is assumed to be zero. Thus, this and the argument of periastron have been omitted from the table.

Table 3: Comparison of this work with Martínez-Palomera et al. (2023) "Kbonus" light curves.

Host star and KIC aperture		This work i	Kbonus i	This work k	Kbonus k	This work a	Kbonus a
Gaia DR3 2100325529865507712	Value	1.4176	1.4205	0.0887	0.0915	5.3011	5.3737
(KIC 4543171)	Std. dev.	0.0044	0.0030	0.0015	0.0010	0.0880	0.0596
Gaia DR3 2079369937757906688	Value	1.4744	1.4462	0.0772	0.0676	6.8480	6.0306
(KIC 9040849)	Std. dev.	0.0207	0.0110	0.0025	0.0016	0.5795	0.2739

Notes. For each exoplanet candidate the value for each principal Pytransit-derived parameter is shown along with the standard deviation of the parameter value. The column headings have the following meanings:
This work: values derived from the light curves described in this work.
Kbonus: values derived from the light curves published by Martínez-Palomera et al. (2023), "Kepler Bonus: Light Curves of Kepler Background Sources".
i: orbital inclination.
k: ratio of planet radius to stellar radius.
a: ratio of orbital radius to stellar radius.

Table 4: Zero epochs (BJD_0) and orbital periods.

Host star and KIC aperture		Initial value	Transit model correction	TTV correction	Final value
Gaia DR3 2100980735718445312	BJD_0	167.572824	5.068E-04	-1.272E-03	167.572059
(KIC 4459924)	Period	4.7917749		-4.340E-07	4.7917745
Gaia DR3 2100325529865507712	BJD_0	168.033975	3.897E-04	-2.475E-04	168.034117
(KIC 4543171)	Period	3.2630129		-1.577E-06	3.2630113
Gaia DR3 2078315162505745920	BJD_0	349.939350	-3.808E-04	1.354E-04	349.939104
(KIC 8175131)	Period	4.2712062		1.796E-06	4.2712080
Gaia DR3 2126910621514995968	BJD_0	259.582238	-9.730E-05	1.088E-04	259.582020
(KIC 8293539)	Period	3.6228904		2.850E-07	3.6228907
Gaia DR3 2079369937757906688	BJD_0	349.026237	-8.238E-04	-5.544E-04	349.024859
(KIC 9040849)	Period	4.1187603		-9.867E-07	4.1187593
Gaia DR3 2085335681690420608	BJD_0	261.905585	2.050E-04	9.378E-05	261.905884
(KIC 9674230)	Period	3.5465873		-1.140E-08	3.5465873
Gaia DR3 2128959007378441472	BJD_0	170.954020	2.718E-05	-1.424E-05	170.954033
(KIC 11033410)	Period	2.5089134		1.783E-07	2.5089136

Notes. This table shows the values of the zero epoch (BJD_0) and the orbital period for each candidate. The initial value is shown, i.e. the value of the orbital period obtained by PDM and the BJD_0 value obtained from the quadratic fit to the folded light curve. The corrections arising from the Transit modelling (for BJD_0 only) and the TTV analysis are shown and the final corrected values. BJD_0 is shown as $BJD - 2454833$ days and the period figures are all in days.

Table 5: Host star parameters.

Host star and KIC aperture	Value	m_G	T_{eff} (K)	log(g) (log(cm s^{-2}))	Distance (pc)	M_G	Luminosity ($L_\odot$)	Radius ($R_\odot$)	Mass ($M_\odot$)
Gaia DR3 2100980735718445312	low		5211	4.269	1853.7	4.3469	1.3591	1.4303	1.3882
(KIC 4459924)	mean	15.6871	5240	4.286	1982.1	4.2015	1.5539	1.5125	1.6143
	high		5273	4.303	2100.4	4.0756	1.7449	1.5828	1.8384
Gaia DR3 2100325529865507712	low		5926	4.158	1740.7	3.7797	2.2915	1.4361	1.0839
(KIC 4543171)	mean	14.9833	5977	4.186	1789.4	3.7198	2.4216	1.4512	1.1805
	high		6071	4.210	1849.2	3.6484	2.5861	1.4536	1.2517
Gaia DR3 2078315162505745920	low		5245	4.609	2292.7	5.0335	0.7221	1.0291	1.5723
(KIC 8175131)	mean	16.8352	5289	4.627	2544.2	4.8074	0.8892	1.1231	1.9518
	high		5344	4.650	2852.5	4.5591	1.1178	1.2334	2.4821
Gaia DR3 2126910621514995968	low		5408	4.625	2148.0	4.8890	0.8249	1.0346	1.6487
(KIC 8293539)	mean	16.5492	5476	4.644	2362.0	4.6828	0.9974	1.1096	1.9812
	high		5543	4.662	2558.4	4.5094	1.1702	1.1730	2.3077
Gaia DR3 2079369937757906688	low		5548	4.335	1803.1	4.6478	1.0301	1.0985	0.9533
(KIC 9040849)	mean	15.9279	5569	4.360	1909.5	4.5233	1.1553	1.1546	1.1155
	high		5588	4.382	2015.3	4.4062	1.2868	1.2103	1.2894
Gaia DR3 2085335681690420608	low		5545	4.252	1259.0	4.4997	1.1807	1.1774	0.9045
(KIC 9674230)	mean	14.9998	5579	4.264	1290.1	4.4467	1.2397	1.1918	0.9528
	high		5614	4.276	1330.0	4.3805	1.3176	1.2134	1.0153
Gaia DR3 2128959007378441472	low		5096	4.457	2742.2	5.3448	0.5421	0.9446	0.9334
(KIC 11033410)	mean	17.5353	5119	4.590	3271.3	4.9617	0.7715	1.1167	1.7721
	high		5161	4.643	4493.2	4.2725	1.4554	1.5090	3.6556

Notes. This table shows the host star data taken from the Gaia catalogue: apparent magnitude, m_G, effective temperature, T_{eff} and gravity, $log(g)$ together with the photogeometric distances to the stars calculated by **?**. From this data, we have calculated the values of the absolute magnitude, M_G, luminosity, radius, and mass. The stellar radii and masses are used to calculate the values of the planetary radii and orbital radii shown in Table 6.
The Gaia catalogue and Bailer-Jones distance catalogue show low/mean/high values rather than ± errors and, for consistency, we have retained this format.

Table 6: Exoplanet parameters.

Host star and KIC aperture	Value	BJD_0 (BJD -2454833)	Period (Days)	i (Rad.)	Orb. rad. (1) (A.U.)	a	Orb. rad. (2) (A.U.)	k	Planet rad. (R_{Jup})
Gaia DR3 2100980735718445312	Low			1.4511	0.0621	5.8419	0.0389	0.0737	1.0500
(KIC 4459924)	Mean	167.572059	4.7917745	1.4614	0.0653	6.0611	0.0427	0.0752	1.1329
	High			1.4717	0.0681	6.2768	0.0462	0.0767	1.2092
Gaia DR3 2100325529865507712	Low			1.4132	0.0442	5.2131	0.0348	0.0872	1.2473
(KIC 4543171)	Mean	168.034117	3.2630113	1.4176	0.0455	5.3011	0.0358	0.0887	1.2821
	High			1.4221	0.0464	5.3891	0.0365	0.0902	1.3060
Gaia DR3 2078315162505745920	Low			1.4938	0.0599	8.9195	0.0427	0.1306	1.3387
(KIC 8175131)	Mean	349.939104	4.2712080	1.5060	0.0644	9.4639	0.0495	0.1349	1.5091
	High			1.5183	0.0698	10.0083	0.0575	0.1392	1.7101
Gaia DR3 2126910621514995968	Low			1.4826	0.0545	7.6237	0.0367	0.1355	1.3963
(KIC 8293539)	Mean	259.582238	3.6228907	1.4872	0.0580	7.7727	0.0401	0.1373	1.5174
	High			1.4918	0.0610	7.9217	0.0433	0.1391	1.6252
Gaia DR3 2079369937757906688	Low			1.4536	0.0495	6.2685	0.0321	0.0747	0.8174
(KIC 9040849)	Mean	349.024859	4.1187593	1.4744	0.0522	6.8480	0.0368	0.0772	0.8878
	High			1.4951	0.0547	7.4275	0.0418	0.0797	0.9608
Gaia DR3 2085335681690420608	Low			1.4403	0.0440	6.4500	0.0353	0.0999	1.1715
(KIC 9674230)	Mean	261.905884	3.5465873	1.4565	0.0448	7.0327	0.0390	0.1037	1.2310
	High			1.4726	0.0457	7.6154	0.0430	0.1075	1.2992
Gaia DR3 2128959007378441472	Low			1.4268	0.0353	5.5210	0.0243	0.1127	1.0603
(KIC 11033410)	Mean	170.954033	2.5089136	1.4328	0.0437	5.6436	0.0293	0.1146	1.2747
	High			1.4388	0.0557	5.7662	0.0405	0.1165	1.7510

Notes. The parameters shown for each candidate planet are:
BJD_0: the zero epoch.
Period: the orbital period of the candidate.
i: the orbital inclination obtained from PyTransit modelling.
Orb. rad. (1): The orbital radius calculated from the period and the stellar mass.
a: the ratio of orbital radius to stellar radius obtained from PyTransit modelling.
Orb. rad. (2): the orbital radius calculated from a and the stellar radius.
k: the ratio of planet radius to stellar radius obtained from PyTransit modelling.
Planet rad.: the radius of the planet calculated from k and the stellar radius.
The high and low values of i, a and k are the mean values ± one standard deviation.
Note that the orbital eccentricity is zero in all cases.
As in Table 5, for clarity and consistency, we present the data in low/mean/high format rather than with ± errors.

Table 7: Odd and even transit model results.

Host star and KIC aperture	Transit depth	1σ uncertainty	Odd transit depth	Odd depth difference/σ	Even transit depth	Even depth difference/σ
Gaia DR3 2100980735718445312 (KIC 4459924)	0.006190	0.000247	0.006202	0.048	0.006192	0.009
Gaia DR3 2100325529865507712 (KIC 4543171)	0.007526	0.000255	0.007507	-0.072	0.007467	-0.232
Gaia DR3 2078315162505745920 (KIC 8175131)	0.020316	0.001296	0.020340	0.019	0.020168	-0.114
Gaia DR3 2126910621514995968 (KIC 8293539)	0.020742	0.000529	0.020893	0.284	0.020460	-0.533
Gaia DR3 2079369937757906688 (KIC 9040849)	0.006520	0.000423	0.006476	-0.105	0.006620	0.240
Gaia DR3 2085335681690420608 (KIC 9674230)	0.010088	0.000730	0.010280	0.264	0.010119	0.042
Gaia DR3 2128959007378441472 (KIC 11033410)	0.012948	0.000418	0.012925	-0.054	0.012842	-0.253

Notes. The values shown for each candidate planet are:
Transit depth: the full light curve fractional transit depth obtained from the Pytransit model.
1σ uncertainty: the 1 standard deviation uncertainty in the Transit Depth.
Odd transit depth: the transit depth of the the odd-numbered transits obtained from the Pytransit model.
Odd depth difference/σ: the difference between the full light curve transit depth and the odd-numbered transits depth expressed as a fraction of the full transit uncertainty.
Even transit depth: the transit depth of the the even-numbered transits obtained from the Pytransit model.
Even depth difference/σ: the difference between the full light curve transit depth and the even-numbered transits depth expressed as a fraction of the full transit uncertainty.